VHAWPB BAS Facility Standards

Veteran Affairs Healthcare System
7305 North Military Trail
West Palm Beach, FL. 33410

1. General Notes

1.1. The standards listed in this document are to be adhered to by all architectural engineers, contractors, vendors, and facility personnel during design, implementation, and maintenance of systems, equipment, software, programming, and all ancillary items on the Building Automation System (BAS).

1.2. These standards shall be in coordination with the most recent standards, codes and guidelines listed in the Technical Information library (TIL) which include but are not limited to:

- 1.2.1. The TIL can be found by the following:
  - 1.2.1.1. https://www.cfm.va.gov/TIL/
- 1.2.2. VA access for the following codes can be found through CEOSH:
  - 1.2.2.1. http://vaww.hefp.va.gov/resources/standards-codes-library
- 1.2.3. HVAC Design Manual
  - 1.2.3.1. https://www.cfm.va.gov/til/dManual/dmHVAC.pdf
- 1.2.4. Section 23 09 23: Digital Control Systems for HVAC
  - 1.2.4.1. https://www.cfm.va.gov/TIL/spec/230923.docx
- 1.2.5. ANSI/ASHRAE/ASHE 135: BACnet – A Data Communication Protocol for Building Automation and Control Networks
- 1.2.6. ANSI/ASHRAE/ASHE 170: Ventilation of Health Care Facilities
- 1.2.7. ANSI/TIA-862: Building Automation Systems Cabling Standard
- 1.2.8. ASHRAE Handbook: Heating, Ventilating, and Air-Conditioning Applications
- 1.2.9. NFPA 70: National Electric Code
- 1.2.10. NFPA 99: Healthcare Facilities Code
- 1.2.11. NFPA 110: Standard for Emergency and Standby Power Systems
- 1.2.12. FGI Guidelines for Design and Construction of Hospitals

2. Building Automation System (BAS) Summary

2.1. All items listed under section 2 in this document shall be incorporated with the existing installed BAS in accordance with the TIL and standards listed in this document to include but not limited to all point programming, graphic display, physical connections, and associated labeling.

2.2. Multiple BAS systems may be utilized by the facility at the time of system modifications. Any changes made to the system must be mirrored on each alternate active BAS.

2.3. Items not listed in this section but considered essential for integration with the BAS are at the facilities discretion and may be included in contracting requirements, service requests, or work orders.

2.4. Equipment direct digital controls shall utilize existing BACnet communication and/or IP networks.

2.5. Mechanical
- 2.5.1. Air Handler Unit
- 2.5.2. Roof Top Unit
- 2.5.3. CAV Unit
- 2.5.4. VAV Unit
- 2.5.5. Temperature Sensor
- 2.5.6. Humidity Sensor
- 2.5.7. Room Pressure Monitor
- 2.5.8. Exhaust Fan
- 2.5.9. Cooling Tower
- 2.5.10. Damper
- 2.5.11. Vent
- 2.5.12. Refrigerator
- 2.5.13. Freezer
- 2.5.14. Hood
- 2.5.15. Air Filter
- 2.5.16. Air Compressor
- 2.5.17. Humidifier – (located at the end of the AHU graphic before the temp/humidity sensor)

2.6. Electrical
- 2.6.1. VFD
- 2.6.2. Motor
- 2.6.3. Substation
- 2.6.4. Paralleling Gear
- 2.6.5. Generator
- 2.6.6. UPS
- 2.6.7. Photocell
- 2.6.8. Exterior Lighting Circuit
- 2.6.9. Elevator
- 2.6.10. Utility Meter
- 2.6.11. Traffic Gates

2.7. Plumbing
- 2.7.1. Pump
- 2.7.2. Water Tower

2.7.3. Valve
2.7.4. Temperature Sensor
2.7.5. Pressure Sensor
2.7.6. Exchanger
2.7.7. Chiller
2.7.8. Medgas
2.7.9. Elevator sump pump

2.8. Infrastructure
2.8.1. Door Status Sensor

2.9. Life Safety
2.9.1. Duct Smoke Detectors
2.9.2. Building Service Equipment Lockout

2.10. Critical Equipment
2.10.1. AHU'S labeled as critical (currently AHU's – 2, 4, 20, 21, 22, 24, 25, 27, 28, 60, and 64). Verify critical AHU's with facility prior to each project and design.
2.10.2. Chillers
2.10.3. Cooling towers
2.10.4. Reheat loop
2.10.5. Air compressors
2.10.6. Water tower
2.10.7. Water heaters
2.10.8. All points listed on the critical data graphics screen or as otherwise designated by the facility during design.

2.11. System
2.11.1. User Profiles (Privileges)
2.11.1.1. Operators (System monitoring, point commands, alarm monitoring, acknowledgement and alarm reset)
2.11.1.2. Energy Center Supervisor (Admin)
2.11.1.3. M&O Chief (Admin)
2.11.1.4. HVAC Supervisor (System monitoring, point commands, and alarm monitoring)
2.11.1.5. HVAC mechanics – 6 (System monitoring, and alarm monitoring)
2.11.1.6. BAS technicians – 2 (Admin)
2.11.1.7. BAS Vendor accounts – account for associated vendors (Admin)
2.11.2. Schedules
2.11.2.1. Time of Day
2.11.2.2. Occupancy
2.11.3. Reports
2.11.3.1. Point summary
2.11.3.1.1. Point Name
2.11.3.1.2. Point Description
2.11.3.1.3. Point Type
2.11.3.1.4. Point Value

    - 2.11.3.1.5. Command Status
    - 2.11.3.1.6. Panel Name
    - 2.11.3.1.7. FLN Number
    - 2.11.3.1.8. Point Address
    - 2.11.3.1.9. Time online
- 2.11.3.2. Points not in auto
    - 2.11.3.2.1. Filtered by command mode
    - 2.11.3.2.2. Point name
    - 2.11.3.2.3. Point description
    - 2.11.3.2.4. Point panel
    - 2.11.3.2.5. Point value
    - 2.11.3.2.6. Point auto setpoint (if applicable)
    - 2.11.3.2.7. Point memo/notes
    - 2.11.3.2.8. Time and date point taken out of auto
    - 2.11.3.2.9. User which removed point from auto
- 2.11.3.3. Point Usage
    - 2.11.3.3.1. Filtered by point name and/or panel.
    - 2.11.3.3.2. Program names using associated points
    - 2.11.3.3.3. Program lines using associated points (including program line #)
    - 2.11.3.3.4. Associated point values
    - 2.11.3.3.5. Associated point names
    - 2.11.3.3.6. Associated point command status
    - 2.11.3.3.7. Associated point description
    - 2.11.3.3.8. Associated point panel
- 2.11.3.4. Alarm/Event/Fault history
    - 2.11.3.4.1. Event classification (Alarm, event, fault, etc.)
    - 2.11.3.4.2. Time and date became active
    - 2.11.3.4.3. Time and date acknowledged
    - 2.11.3.4.4. Time and date cleared
    - 2.11.3.4.5. Time and date reset
    - 2.11.3.4.6. Associated user profile actions
    - 2.11.3.4.7. Alarm/Event/Fault description
    - 2.11.3.4.8. Associated point
    - 2.11.3.4.9. Point Value
    - 2.11.3.4.10. Associated alarm conditions
    - 2.11.3.4.11. Associated panel
    - 2.11.3.4.12. Event ID
- 2.11.3.5. Active Alarms/Events/Faults
    - 2.11.3.5.1. Time and date became active
    - 2.11.3.5.2. Time and date acknowledged
    - 2.11.3.5.3. Associated user profile actions
    - 2.11.3.5.4. Alarm/Event/Fault description

2.11.3.5.5. Associated point
2.11.3.5.6. Point Value
2.11.3.5.7. Associated alarm conditions
2.11.3.5.8. Associated panel
2.11.3.5.9. Event ID

2.11.3.6. Equipment network information

2.11.3.6.1. Equipment name
2.11.3.6.2. Equipment description
2.11.3.6.3. Associated panel
2.11.3.6.4. IP address
2.11.3.6.5. MAC address
2.11.3.6.6. Instance number

2.11.3.7. Trends

2.11.3.7.1. Trends must maintain records for 120 days
2.11.3.7.2. Trends must include any alarm conditions for associated values
2.11.3.7.3. All AHU and critical equipment start/stop or on/off
2.11.3.7.4. All AHU supply air temp
2.11.3.7.5. All AHU supply airflow
2.11.3.7.6. All AHU supply static pressure
2.11.3.7.7. Critical equipment water temp(s)
2.11.3.7.8. Critical equipment water volume(s)
2.11.3.7.9. Critical equipment waterflow(s)
2.11.3.7.10. Critical equipment damper(s) position
2.11.3.7.11. Critical equipment airflow(s)
2.11.3.7.12. Critical equipment air temp(s)
2.11.3.7.13. Critical equipment humidity(s)
2.11.3.7.14. Critical equipment pressure(s)
2.11.3.7.15. Critical equipment valve position(s)
2.11.3.7.16. Critical equipment kW(s)
2.11.3.7.17. Critical equipment current(s)
2.11.3.7.18. Critical equipment frequency(s)
2.11.3.7.19. Critical room temp
2.11.3.7.20. Critical room humidity
2.11.3.7.21. Critical room differential pressure

3. Facility Standards

3.1. Design

3.1.1. Installation, modification, and demolition to equipment with integration to the BAS must be submitted to, evaluated and approved by the facilities engineering department in accordance with the facilities design requirements prior to project work initiation.

3.1.2. In addition to any engineering design requirements listed in the TIL, or other otherwise stated in the contract, design documentation must be in accordance with items listed under Documents section 3.5 in this standards list.

3.1.3. Prior to completion of design, existing conditions must be evaluated for deficiencies to the TIL and facility standards. Any deficiencies found must be formally documented and submitted to the facility engineering department for review, evaluation, and determination for inclusion in the project.

3.1.4. Hardware

- 3.1.4.1. Equipment must be rated for the environmental conditions of the area selected for installation.
- 3.1.4.2. Equipment must have proper utility ratings for installation configurations being utilized.
- 3.1.4.3. Chillers, Boilers, and pumps shall require automatic changeover to the backup or lag piece without facility personnel action should one piece of equipment fail.
- 3.1.4.4. Designs shall include a 20% expansion capability for newly installed or modified equipment including but not limited to field panels, controllers, networking equipment, and UPS's.
- 3.1.4.5. Critical equipment must have power redundancy via UPS or alternatively approved means.
- 3.1.4.6. UPS's must be double conversion and capable of integrating into the facilities UPS monitoring system.
- 3.1.4.7. Upon loss of power or equipment shutdown, and after conditions are naturally or automatically corrected, equipment shall return to normal operating condition without the need for facility personnel actions.

3.1.5. Graphics

- 3.1.5.1. Any contractor, vendor, or facility personnel making a modification to the BAS or associated equipment are responsible for ensuring that all associated points are properly configured into the system and programmed onto the appropriate graphics screen.
- 3.1.5.2. All graphics pages shall be linked to associated graphics pages from equipment and points listed on that page.
- 3.1.5.3. Graphics pages shall be created and organized based on the following structure:
  - 3.1.5.3.1. Equipment
    - 3.1.5.3.1.1. Customizable 2D layout of equipment and all ancillary valves, pumps, motors, sensors, meters, air separators, and piping.
    - 3.1.5.3.1.2. Customizable adjacent graphic to each point with actual values, setpoints, and alarm indicators.
    - 3.1.5.3.1.3. Customizable table with all associated point actual values, setpoints, alarm indicators, and engineering design values.
    - 3.1.5.3.1.4. Graphical icons with links to O&M manual for associated equipment, and/or sequence of operation document.
    - 3.1.5.3.1.5. Equipment graphics pages include but are not limited to the following:
      - 3.1.5.3.1.5.1. AHU

3.1.5.3.1.5.2. VAV/CAV
3.1.5.3.1.5.3. Chiller Plant
3.1.5.3.1.5.4. Chiller
3.1.5.3.1.5.5. Pump House Tank
3.1.5.3.1.5.6. Domestic Hot Water Heater
3.1.5.3.1.5.7. Dietetic Hot Water Heater
3.1.5.3.1.5.8. Domestic Water Tower
3.1.5.3.1.5.9. Cooling Towers
3.1.5.3.1.5.10. Chiller Staging
3.1.5.3.1.5.11. Primary Pump Staging
3.1.5.3.1.5.12. Reheat Loop
3.1.5.3.1.5.13. Angio Chiller
3.1.5.3.1.5.14. AEC1/AEC2
3.1.5.3.1.5.15. Control Air Compressors
3.1.5.3.1.5.16. Street Lights
3.1.5.3.1.5.17. Cook Chilled Water System
3.1.5.3.1.5.18. Electrical Distribution
3.1.5.3.1.5.19. Battery Levels

3.1.5.3.2. Dashboards

3.1.5.3.2.1. Dashboards shall include customizable tables for all associated equipment points actual values, setpoints, alarm indicators and engineering design values.

3.1.5.3.2.2. Any equipment and points matching the description of the dashboard shall be included on the graphics page. Graphics pages not clearly defined in this section will be included at the discretion of the facility engineering department.

3.1.5.3.2.3. Tables will be separated by equipment if possible.

3.1.5.3.2.4. Tables will be separated by mechanical, electrical, and plumbing if possible.

3.1.5.3.2.5. Dashboard graphic pages include but are not limited to the following:

3.1.5.3.2.5.1. Critical Data (Discretion of facility)
3.1.5.3.2.5.2. Power Plant
3.1.5.3.2.5.3. Chill Water Valves
3.1.5.3.2.5.4. Min Dampers
3.1.5.3.2.5.5. Exhaust Fans
3.1.5.3.2.5.6. AHU Time of Day
3.1.5.3.2.5.7. Elevator Status
3.1.5.3.2.5.8. Elevator Vent Dampers
3.1.5.3.2.5.9. AHU Cooling Setpoints
3.1.5.3.2.5.10. AHU Occupied/Unoccupied
3.1.5.3.2.5.11. Hood Vent Damper
3.1.5.3.2.5.12. Fire Vent Dampers

3.1.5.3.2.5.13. AHU Smoke Alarm
3.1.5.3.2.5.14. Refrigeration
3.1.5.3.2.5.15. CLC Refrigeration
3.1.5.3.2.5.16. Basement Refrigeration
3.1.5.3.2.5.17. 1st Floor Refrigeration
3.1.5.3.2.5.18. 4th Floor Refrigeration
3.1.5.3.2.5.19. Kitchen Refrigeration
3.1.5.3.2.5.20. Pharmacy Refrigeration
3.1.5.3.2.5.21. Isolation Rooms
3.1.5.3.2.5.22. Surgical Procedure Room sensors
3.1.5.3.2.5.23. Door Alarms
3.1.5.3.2.5.24. Critical Alarm Panels
3.1.5.3.2.5.25. Critical Bell (must be programmed into critical bell program) (At discretion of facility)
3.1.5.3.2.5.26. VAV (for each floor. Separated by AHU)

3.1.5.3.3. Campus
3.1.5.3.3.1. Customizable background of Campus Site Plan
3.1.5.3.3.2. Graphic point for each building’s temp and humidity layered over building location. Point links to buildings graphic page.
3.1.5.3.3.3. Table displaying each building’s active values and setpoints for temp and humidity.
3.1.5.3.3.4. Traffic gates point actual status, and alarm indicator.

3.1.5.3.4. Main Building
3.1.5.3.4.1. Customizable 2D background of main building sectioned by floor with names of services in each section on each floor.
3.1.5.3.4.2. AHU Icons layered over associated floors and sections with link to associated AHU graphics page.
3.1.5.3.4.3. Table with name and link to each dashboard. Alarm indication associated for each dashboard option. Indicator must separate between acknowledged and unacknowledged alarms. (ex: flashing, color difference, etc.)
3.1.5.3.4.4. Table with name and link to each equipment graphics page. Alarm indication associated for each equipment graphics page option. Indicator must separate between acknowledged and unacknowledged alarms.

3.1.5.3.5. Outer Buildings
3.1.5.3.5.1. Primary floor plan for building. Customizable room sizes and room numbers.
3.1.5.3.5.2. Active value’s and setpoints for temperature and humidity layered over each associated room
3.1.5.3.5.3. Graphical link to associated mechanical graphics page
3.1.5.3.5.4. Graphical link to other floor plan for building

3.1.5.3.5.5. Graphical Icon for the building's active value and setpoint for temperature and humidity

3.1.5.3.6. Floor plan

3.1.5.3.6.1. Customizable room sizes and layout for future projects.

3.1.5.3.6.2. Customizable room numbers

3.1.5.3.6.3. Active value's and setpoints for temperature and humidity layered over each associated room. Linked to associated VAV graphics page.

3.1.5.3.6.4. Graphical icon linked to associated mechanical graphics page.

3.1.5.3.6.5. Table with associated room active values, setpoints, and alarm icons. Linked to associated VAV graphics page.

3.1.5.3.7. Mechanical graphics page

3.1.5.3.7.1. Customizable floor plan based on mechanical drawings with graphics for ducts, VAV's, CAV's, thermostats, etc.

3.1.5.3.7.2. Graphics for equipment linked to associated graphics page.

3.1.5.3.7.3. Table with associated room active values, setpoints, and alarm icons. Linked to associated VAV graphics page.

3.1.5.4. Graphic points shall be created and added to the graphics page for each associated equipment.

3.1.5.5. Commands shall be capable of being issued from graphics points if the associated points have control capabilities. Points which have no controls capabilities shall only display values and not be capable of commanding value changes on graphics.

3.1.5.6. A points list shall be provided to the facility upon completion of the project. Graphics pages and points shall be verified by facility personnel for accuracy and completeness.

3.1.6. Events

3.1.6.1. Monitoring

3.1.6.1.1. Alarms must be created for each point based on engineering design.

3.1.6.1.2. Alarm condition parameters must meet or exceed code requirements.

3.1.6.1.3. Alarms must be able to be configured and edited in the BAS.

3.1.6.1.4. The BAS must monitor changes to alarm conditions and keep logs on those changes.

3.1.6.1.5. Fault alerts must activate if equipment is not communicating.

3.1.6.1.6. Status alerts must activate if equipment needs software update or equipment has returned from failure.

3.1.6.1.7. Alarms must be categorized using the following structure:

3.1.6.1.7.1. Fire Alarm

3.1.6.1.7.1.1. Reserved for Smoke Detectors. Verify Fire alarm system has correlated trouble. If not, notify Electronics shop.

3.1.6.1.7.2. High

3.1.6.1.7.2.1. Requires immediate action to prevent imminent failure of equipment.

3.1.6.1.7.3. Medium

3.1.6.1.7.3.1. Requires immediate acknowledgement. Action may or may not require immediate action depending on multiple variables.

3.1.6.1.7.4. Low

3.1.6.1.7.4.1. Typically preventive maintenance and average priority tasks. Acknowledge and submit work order to associated shops.

3.1.6.1.7.5. Fault

3.1.6.1.7.5.1. Communication signal issues. Requires immediate acknowledgement. Action may or may not require immediate action depending on multiple variables.

3.1.6.1.7.6. Status

3.1.6.1.7.6.1. System quality of life notification. Typically requires work order submitted during normal business hours to building automation team for system updates.

3.1.7. Labeling

3.1.7.1. Naming conventions

3.1.7.1.1. BAS points - Must follow the facility designated naming convention of: Building number_Room number_Equipment_Point type. If a field is not available it shall used a value of 0.

3.1.7.1.2. BAS Panels – Must include primary equipment names, panel number, and IP address on enclosure. Panels in software must include building number, room number, equipment names, and panel number.

3.1.7.1.3. BAS equipment – Must follow existing industry standards

3.1.7.1.4. Abbreviations and suffix's used shall comply with the pre-defined database provided by the facility.

3.1.7.2. Point labeling

3.1.7.2.1. Field panel designations shall be updated with the most recent point names, addresses and descriptions.

3.1.7.2.2. Instance numbers must follow the facilities designated numbering scheme of "S BB PP DD".

3.1.7.2.2.1. S = Site Designator (1-4)

3.1.7.2.2.2. BB = Building Designator

3.1.7.2.2.3. PP = Panel Number

3.1.7.2.2.4. DD = Device MAC address (00 is reserved for the panel)

3.1.7.2.2.5. All instance numbers must be unique

3.1.7.2.2.6. MAC address for a BACnet IP device is the IP address combined with the port number used

3.1.7.2.2.7. MAC address for and MSTP device is the drop number

3.1.7.3. Equipment labeling

3.1.7.3.1. Equipment shall be labeled with equipment ID, Area being serviced, and associated panel ID.

3.1.7.3.2. Labelling shall not be obstructed in any way and be easily visible in front of equipment.

3.1.7.4. Cable labeling

3.1.7.4.1. All cables must be labeled on both ends.

3.1.7.4.2. Cables on equipment/point side must be labeled with the panel ID.

3.1.7.4.3. Cables on panel side must be labeled with point name.

3.1.8. Reporting

3.1.8.1. Events and Alarms

3.1.8.1.1. Alarm, event, and fault history

3.1.8.1.2. Active alarms, events, and faults

3.1.8.1.3. Point usage

3.1.8.1.4. Point summary

3.1.8.1.5. Points not in auto

3.1.8.1.6. Equipment network information

3.1.9. Network and Cabling

3.1.9.1. Networking

3.1.9.1.1. Equipment's primary communication must be hard wired. Wireless may only be permitted if hard wired is not available and must be approved by the facility engineering department prior to design completion and installation.

3.1.9.1.2. Networking information must not conflict with existing network infrastructure. Verification must be made with the facilities equipment inventory.

3.1.9.1.3. Upon network failure equipment shall operate locally and return to last known good configuration or value if possible.

3.1.9.1.4. The headend switch must contain the following minimum specs.

3.1.9.1.4.1. Layer 3

3.1.9.1.4.2. 10 Gbps or greater SFP+ ports to accommodate existing and future planned project port needs with an additional 4 spare ports

3.1.9.1.4.3. SFP to LC fiber transceiver modules for all fiber connections

3.1.9.1.4.4. OM5 or greater fiber optic cable (or newest industry standard)

3.1.9.1.4.5. LC fiber optic terminal connectors

3.1.9.1.4.6. SSL3 and TLS 2.0 encryption

3.1.9.1.4.7. Authenitcation Radius, SNMPv3, TACACS+, Secure shell v.2 (SSH2), and webGUI management

3.1.9.1.4.8. Features: Power auto-reboot, rapid spanning tree, multiple spanning tree, VLAN assignment, Port trunking

3.1.9.1.5. Switches other than the headend must contain the following minimum specs.

3.1.9.1.5.1. Layer 2

3.1.9.1.5.2. 10 Gbps or greater RJ-45 ethernet ports to accommodate existing and future planned project port needs with an additional 4 spare ports.

3.1.9.1.5.3. At least 2 x 10GBase-X SFP+

3.1.9.1.5.4. SFP to LC Fiber Transceiver Modules for SFP ports

3.1.9.1.5.5. Encryption: TLS 2.0

3.1.9.1.5.6. POE+ 30W per port

3.1.9.1.5.7. Authentication: RADIUS, SNMPv3, TACACS+, Secure Shell v.2 (SSH2), and WebGUI management

3.1.9.1.5.8. Warranty: 5 year

3.1.9.1.5.9. Feature: Power Auto-reboot, Rapid Spanning Tree (RSTP) and Multiple Spanning Tree (MSTP), VLAN assignment, Port trunking

3.1.9.1.5.10. Cat6a cable (or newest compatible industry standard)

3.1.9.1.5.11. RJ 45 connectors

3.1.9.1.5.12. OM5 or greater fiber optic cable

3.1.9.1.5.13. LC Fiber Optic terminal connectors

3.1.9.1.5.14. All cabling must be certified

3.1.9.2. Cabling

3.1.9.2.1. Fiber optic cable must be OM-5 or greater

3.1.9.2.2. Fiber connections must be LC

3.1.9.2.3. Ethernet cable must be Cat6A or greater

3.1.9.2.4. Ethernet cable must be plenum rated

3.1.9.2.5. All cabling must be certified prior to project completion

3.1.9.3. Workstations

3.1.9.3.1. BAS workstations shall be setup with the necessary software, programming, networking configurations, and profiles for the purpose of monitoring, maintaining, or managing the existing building automation systems being utilized at the facility.

3.1.9.3.2. Workstations shall be installed and configured for vendors of associated building automation systems with administrator privileges for the purposes of maintenance and installation.

3.1.9.3.3. Workstations designated for the purposes of monitoring shall have access with view only rights to all building automation systems being utilized by the facility.

3.1.9.3.4. Workstations designated for the purposes of maintenance shall have admin access to all building automation systems being utilized by the facility.

3.1.9.3.5. Workstations designated for the purposes of management shall have access for viewing and command rights to all building automation systems being utilized by the facility.

3.1.9.3.6. Workstations shall be installed at the following locations for the associated purposes:

3.1.9.3.6.1. BE-111 – Energy Center Operators (Managing)
3.1.9.3.6.2. BE-111 – Energy Center Siemens BAS (Maintaining)
3.1.9.3.6.3. BE-111 – Energy Center Tridium BAS (Maintaining)
3.1.9.3.6.4. BE-116 – Energy Center Supervisor (Managing)
3.1.9.3.6.5. BC-242 – HVAC Supervisor (Managing)
3.1.9.3.6.6. BC-242 – HVAC Mechanics (2 Total)(Monitoring)
3.1.9.3.6.7. BC-231G – M&O Chief (Managing)
3.1.9.3.6.8. BE-150 – BAS Technicians (2 Total)(Maintaining)

3.2. Existing conditions

3.2.1. Prior to conducting any demolition, installation, or modification to equipment or systems with integration to BAS, the contractor, vendor, or facility personnel responsible for completing the install or modification must verify existing conditions to associated equipment, points, and graphics on the BAS.

3.2.2. Any deficiencies to codes, VA standards, or facility standards found during the evaluation of existing conditions must be submitted in writing to the facilities engineering department prior to project work initiation.

3.2.3. Deficiencies submitted to the facilities engineering department must be recorded by the engineering department for either immediate or future repair and maintained in the projects folder.

3.2.4. Existing condition deficiencies documented prior to project work initiation will not be the responsibility of the contractor, vendor, or facility personnel conducting the installation or modification unless otherwise stated in the contract, purchase order or work order.

3.2.5. Deficiencies found during existing condition evaluations shall be resolved at the discretion of the facilities engineering department.

3.3. Installation/Modification

3.3.1. Installation, modification, or demolition to equipment with integration to the BAS must be submitted to, evaluated and approved by the facilities engineering department in accordance with the facilities design requirements prior to project work initiation.

3.3.2. All work must be in compliance with the TIL and the facility standards listed in this document.

3.3.3. All controls technicians and vendors shall be certified by Niagara N4, thru any various distributors to include but not limited to Siemens, Trane, Niagara, Honeywell, JCI, and Distech. Certifications must be provided as part of the submittal process and total number of certified technicians.

3.4. Inspection/Training

3.4.1. Commissioning

3.4.1.1. There shall be a minimum of 3 classroom design meetings with stakeholders to ensure that all design intents and Owner Requirements are met. There shall be a minimum of 3 classroom style training sessions that cover all points on screens and the actual functional sequence of all point (what does this setpoint do)

3.4.1.2. There will be a minimum of 2 final training sessions with operators on final install so that all shifts can make it.

3.4.1.3. All graphics and controls shall undergo a FAT or Factory Acceptance Test meaning it will be set up on a mock computer and a simulated run will need to be performed and accepted by the VA (no exceptions)

3.4.1.4. Verify all critical equipment has properly programmed start-up, shut-down, normal run, and re-start programs for operation. Operators shall not have to manually run through these sequences.

3.4.1.5. The contractor shall retain a CITP CxP and this system will be commissioned to include the following:

- 3.4.1.5.1. Design Review
  - 3.4.1.5.1.1. Create an OPR for the complete project
  - 3.4.1.5.1.2. Provide a Cx plan
  - 3.4.1.5.1.3. Review design documents and make comments on discrepancies between the design and the OPR
  - 3.4.1.5.1.4. Offer recommendations on design of system and sequence
- 3.4.1.5.2. Construction Review
  - 3.4.1.5.2.1. Mechanical submittals
  - 3.4.1.5.2.2. Equipment submittals
  - 3.4.1.5.2.3. BAS controls submittals
  - 3.4.1.5.2.4. All sequence of operations
- 3.4.1.5.3. Commissioning Plan
  - 3.4.1.5.3.1. Mechanical startup forms by contractor on:
    - 3.4.1.5.3.1.1. VFD's
    - 3.4.1.5.3.1.2. Pumps
    - 3.4.1.5.3.1.3. Control Valves
    - 3.4.1.5.3.1.4. BAS controls contractor "Point Checklist" of sensors and controls devices and pre-functional testing. Controls Contractors point checklist should include what's bad/damaged/needs replacement/out-of-range/etc.
  - 3.4.1.5.3.2. Functional Performance Test Procedures
    - 3.4.1.5.3.2.1. Test and Balance testing and report review
    - 3.4.1.5.3.2.2. DDC System Review
    - 3.4.1.5.3.2.3. Sensor and feedback control calibration review
    - 3.4.1.5.3.2.4. Pre-Commission Meeting
    - 3.4.1.5.3.2.5. On-site Functional Performance Testing
    - 3.4.1.5.3.2.6. Corrective action report – Deliverable:
      - 3.4.1.5.3.2.6.1. Issues Resolution Meeting x4
      - 3.4.1.5.3.2.6.2. Internal QA/QC
      - 3.4.1.5.3.2.6.3. Final Commissioning Report
  - 3.4.1.5.3.3. Commissioning agent to issue a progress draft within 1 week upon the delivery dates for the final commissioning. The final report date will be due no later than 3 weeks after completion of the draft.

3.4.2. BACnet and IP devices must have communications and controls tested, approved and verified working without any issue for all devices installed, or modified on each project by the facilities engineering department prior to project completion.

3.5. Documentation

3.5.1. All items listed under section 3.1 in this document shall be completed, delivered to, and signed by the facility engineering department during design and upon completion of any project performed by contractors, vendors and facility personnel as applicable.

3.5.2. Any contractor, vendor, or facility personnel making a modification to the BAS or associated equipment are responsible for ensuring the following are completed, updated, delivered to, and signed by the facility engineering department.

- 3.5.2.1. Wire Shark report
- 3.5.2.2. Mechanical drawings
- 3.5.2.3. One-line diagrams on associated BAS controls
- 3.5.2.4. One-line diagrams on associated networking equipment
- 3.5.2.5. Building network infrastructure drawing
- 3.5.2.6. Control schematics
- 3.5.2.7. Sequence of operations
  - 3.5.2.7.1. Overview narrative
  - 3.5.2.7.2. Interactions
  - 3.5.2.7.3. Interlocks
  - 3.5.2.7.4. Delineation of control with packaged controllers
  - 3.5.2.7.5. Startup
  - 3.5.2.7.6. Warmup
  - 3.5.2.7.7. Cool Down
  - 3.5.2.7.8. Occupied
  - 3.5.2.7.9. Unoccupied
  - 3.5.2.7.10. Optimal Start/Stop
  - 3.5.2.7.11. Capacity Control
  - 3.5.2.7.12. Staging
  - 3.5.2.7.13. Set points
  - 3.5.2.7.14. Setbacks
  - 3.5.2.7.15. Setups
  - 3.5.2.7.16. Resets
  - 3.5.2.7.17. Demand limiting
  - 3.5.2.7.18. Loss of network
  - 3.5.2.7.19. Loss of power
  - 3.5.2.7.20. Alarms
  - 3.5.2.7.21. Delays
  - 3.5.2.7.22. Must result in fewer questions, and require only limited adjustment of controls contractor.
  - 3.5.2.7.23. Supplied at 35% design and throughout design process
- 3.5.2.8. Associated controls points list

3.5.2.8.1. Must delineate analog or digital outputs
3.5.2.8.2. Must be based on equipment features, sequence of operation and facilities intent.

3.5.2.9. DDC object list
3.5.2.10. Alarm configurations

3.5.2.10.1. Point Name
3.5.2.10.2. Point Description
3.5.2.10.3. Point address
3.5.2.10.4. Point type
3.5.2.10.5. Panel name
3.5.2.10.6. Panel Instance
3.5.2.10.7. Panel IP address
3.5.2.10.8. Associated primary equipment name
3.5.2.10.9. Point value units
3.5.2.10.10. Point alarm high setpoint
3.5.2.10.11. Point alarm low setpoint
3.5.2.10.12. Point alarm notification settings

3.5.2.11. Inventory information on associated equipment utilizing facility provided spreadsheet. Items include but are not limited to:

3.5.2.11.1. Equipment Alias
3.5.2.11.2. Equipment associated Service (Ex: Laundry)
3.5.2.11.3. Controller ID
3.5.2.11.4. IP Address
3.5.2.11.5. Building number
3.5.2.11.6. Floor number
3.5.2.11.7. Room number
3.5.2.11.8. Chiller Loop
3.5.2.11.9. Project name and number
3.5.2.11.10. Equipment Type (Ex: Single zone AHU, VAV, CAV, VFD, etc.)
3.5.2.11.11. Manufacturer
3.5.2.11.12. Model number
3.5.2.11.13. Serial number
3.5.2.11.14. MAC address
3.5.2.11.15. Instance number
3.5.2.11.16. Engineering design CFM
3.5.2.11.17. Engineering design Current
3.5.2.11.18. Engineering design GMP
3.5.2.11.19. Engineering design air exchange rate

3.5.2.12. Updated trending configurations

3.5.2.13. Updated engineering design specification values and setpoints on associated equipment. Setpoints/Values must not conflict with VA codes, standards, or regulations. Setpoints/Values must account for area, supply, return, heating, cooling, static, differential, day, night, occupied and unoccupied configurations if applicable. Setpoints/Values required include but are not limited to the following.

- 3.5.2.13.1. References:
  - 3.5.2.13.1.1. ASHRAE 170-7: Space Ventilation – Inpatient space setpoints
  - 3.5.2.13.1.2. ASHREA 170-8: Space Ventilation- Outpatient space setpoints
  - 3.5.2.13.1.3. ASHRAE 170-9: Space Ventilation – Resident health, care, and support space setpoints
- 3.5.2.13.2. Mechanical:
  - 3.5.2.13.2.1. Air Volume (CFM or ft3/m)
  - 3.5.2.13.2.2. Air Flow (%)
  - 3.5.2.13.2.3. Alarm Setpoints
  - 3.5.2.13.2.4. Pressure (psi)
  - 3.5.2.13.2.5. Humidity (%)
  - 3.5.2.13.2.6. On/Off
  - 3.5.2.13.2.7. Open/Close (%)
  - 3.5.2.13.2.8. Speed (%)
  - 3.5.2.13.2.9. Static Pressure (psi)
  - 3.5.2.13.2.10. Temperature (F/C)
- 3.5.2.13.3. Electrical:
  - 3.5.2.13.3.1. Alarm Setpoints
  - 3.5.2.13.3.2. Current (A)
  - 3.5.2.13.3.3. Frequency (Hz)
  - 3.5.2.13.3.4. On/Off
  - 3.5.2.13.3.5. Power (kW)
  - 3.5.2.13.3.6. Power Factor
  - 3.5.2.13.3.7. Tonnage (Ton)
  - 3.5.2.13.3.8. Voltage (V)
- 3.5.2.13.4. Plumbing:
  - 3.5.2.13.4.1. Alarm Setpoints
  - 3.5.2.13.4.2. Differential Pressure (psi)
  - 3.5.2.13.4.3. Flow (gpm)
  - 3.5.2.13.4.4. On/Off
  - 3.5.2.13.4.5. Open/Close (%)
  - 3.5.2.13.4.6. Temperature (F)